6 Nyquist diagrams

6.1 Introduction

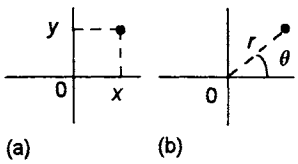

Figure 6.1 *(a) Cartesian graph with points specified by x and y values, (b) polar graph with points specified by r and θ values*

This chapter follows on from Chapter 5 and presents another method of considering the frequency response of systems and their stability. The method uses *Nyquist diagrams*; in these diagrams the gain and the phase of the open-loop transfer function, i.e. the product of the forward path and the feedback path transfer functions, are plotted as polar graphs for various values of frequency. With Cartesian graphs the points are plotted according to their x and y coordinates from the origin; with polar graph the points are plotted from the origin according to their radial distance from it and their angle to the reference axis (Figure 6.1).

6.2 The polar plot

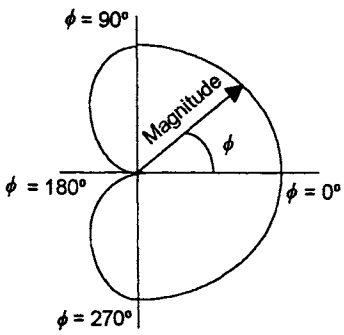

Figure 6.2 *Polar plot with the plot as the line traced out by the tips of the phasors as the frequency is changed from zero to inifnity*

The *polar plot* of the frequency response of a system is the line traced out as the frequency is changed from 0 to infinity by the tips of the phasors whose lengths represent the magnitude, i.e. amplitude gain, of the system and which are drawn at angles corresponding to their phase (Figure 6.2).

Example

Draw the polar diagram for the following frequency response data.

Freq. rad/s	1.4	2.0	2.6	3.2	3.8
Magnitude	1.6	1.0	0.6	0.4	0.2
Phase deg.	−150	−160	−170	−180	−190

Note that the above data gives the phases with negative signs. This means they are lagging behind the 0° line by the amounts given. Figure 6.3 shows the polar plot obtained.

6.2.1 Nyquist diagrams

The term *Nyquist diagram* is used for:

The Nyquist diagram is the line joining the series of points plotted on a polar graph when each point represents the magnitude and phase of the open-loop frequency response corresponding to a particular frequency.

To plot the Nyquist diagram from the open-loop transfer function of a system we need to determine the magnitude and the phase as functions of frequency.

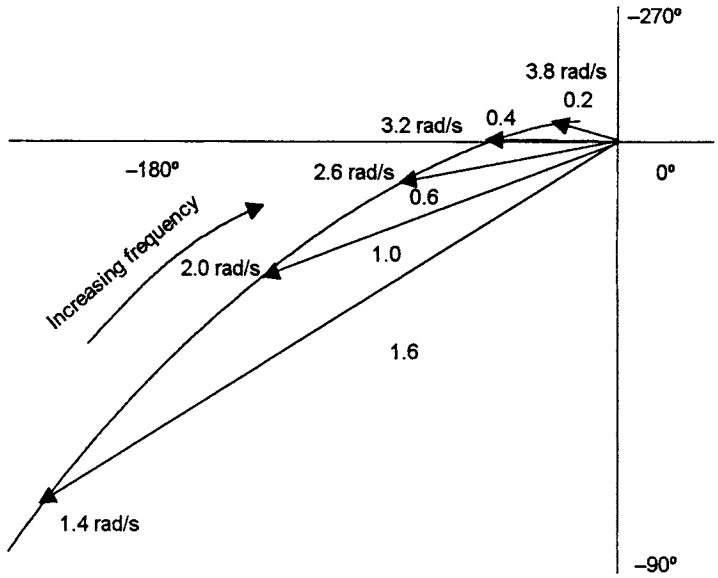

Figure 6.3 *Example*

Example

Determine the Nyquist diagram for a first-order system with an open-loop transfer function of $1/(1+\tau s)$.

The frequency response is:

$$\frac{1}{1+j\omega\tau} = \frac{1}{1+j\omega\tau}\frac{1-j\omega\tau}{1-j\omega\tau} = \frac{1}{1+\omega^2\tau^2} - j\frac{\omega\tau}{1+\omega^2\tau^2}$$

The magnitude is thus:

$$\text{magnitude} = \frac{1}{\sqrt{1+\omega^2\tau^2}}$$

and the phase is:

$$\text{phase} = -\tan^{-1}\omega\tau$$

At zero frequency the magnitude is 1 and the phase 0°. At infinite frequency the magnitude is zero and the phase is –90°. When $\omega\tau = 1$ the magnitude is $1/\sqrt{2}$ and the phase is –45°. Substitution of other values leads to the result shown in Figure 6.4 of a semicircular plot.

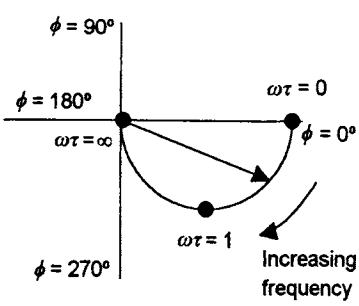

Figure 6.4 *Nyquist diagram for a first-order system*

Example

Determine the Nyquist plot for the system having the open-loop transfer function of $1/s(s+1)$.

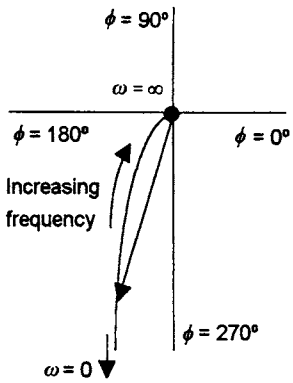

Figure 6.5 *Example*

The frequency response is:

$$G(j\omega) = \frac{1}{j\omega(j\omega+1)} = \frac{1}{j\omega-\omega^2} = \frac{1}{j\omega-\omega^2}\frac{-j\omega-\omega^2}{-j\omega-\omega^2}$$

$$= \frac{-\omega^2}{\omega^2+\omega^4} - j\frac{\omega}{\omega^2+\omega^4}$$

the magnitude and phase are thus:

$$\text{magnitude} = \frac{1}{\omega\sqrt{\omega^2+1}}$$

$$\text{phase} = \tan^{-1} -1/(-\omega) = 180° + \tan^{-1} 1/\omega$$

When $\omega = \infty$ then the magnitude is 0 and the phase is 0°. As ω tends to 0 then the magnitude tends to infinity and the phase to 270° or −90°. Figure 6.5 shows the polar plot.

6.3 Stability

As indicated in Section 5.6: the critical point which separates stable from unstable systems is when the open-loop phase shift is −180° and the open-loop magnitude is 1. If a Nyquist diagram of the open-loop frequency response is plotted then for the system to be stable there must not be any phasor with length greater than 1 and phase −180°. Thus the line traced by the tips of the phasors, the so-termed loci, must not enclose the −1 point.

> *Closed-loop systems whose open-loop frequency response $G(j\omega)H(j\omega)$ loci, as ω goes from 0 to ∞, do not encircle the −1 point will be stable, those which encircle the −1 point are unstable and those which pass through the −1 point are marginally stable. Encircling the point may be taken as passing to the left of the point.*

The above statement is known as the *Nyquist stability criterion*.

Figure 6.6 illustrates the above with examples of stable, marginally stable and unstable systems. The Nyquist plots, not to scale, correspond to the open-loop frequency response of:

$$G(j\omega)H(j\omega) = \frac{K}{(1+j\omega 0.2)(1+j\omega)(1+j\omega 10)}$$

with $K = 10$ for the stable plot, $K = 137$ for the marginally stable plot and $K = 500$ for the unstable plot.

The vertical axis of the Nyquist plot corresponds to the phase equal to 90° and so is the imaginary part of the open-loop frequency response. The horizontal axis corresponds to the phase equal to 0° and so is the real part of the open-loop frequency response.

Figures 6.7 to 6.10 shows examples of Nyquist plots for common forms of open-loop transfer functions and their conditions for stability.

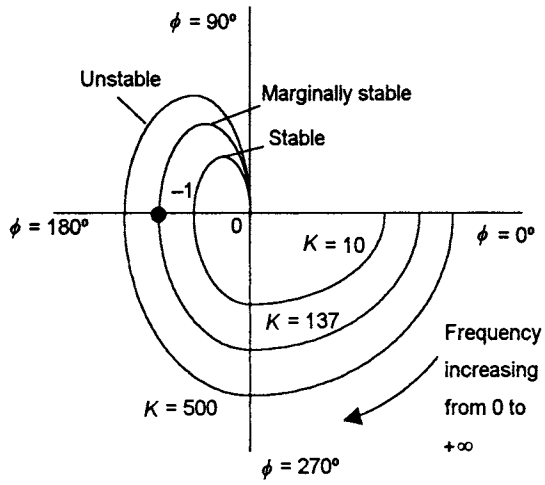

Figure 6.6 *Stability and the Nyquist plot*

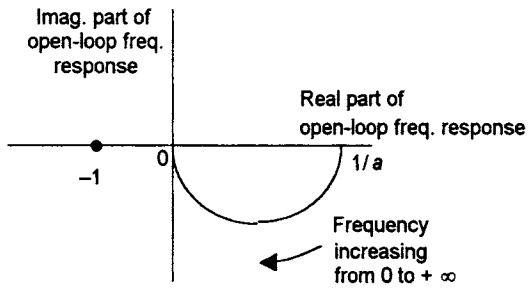

Figure 6.7 *$G(s)H(s) = K/(s + a)$, stable for all values of $K > 0$*

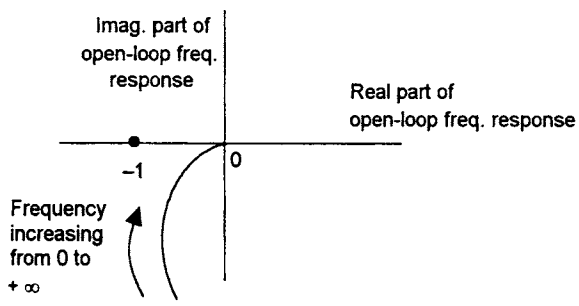

Figure 6.8 *$G(s)G(s) = K/s(s + a)$, stable for all values of $K > 0$*

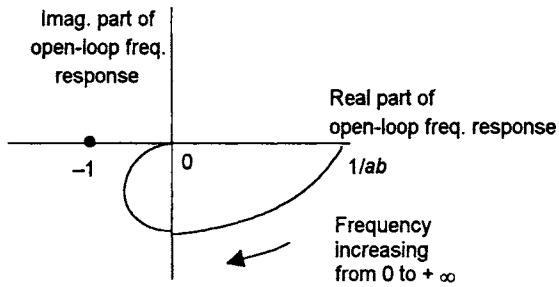

Figure 6.9 *G(s)H(s) = K/(s + a)(s + b), stable for all values of K > 0*

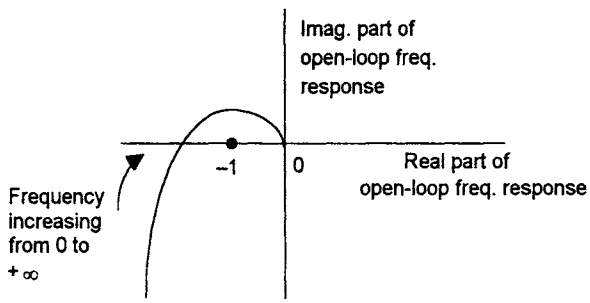

Figure 6.10 *G(s)H(s) = K/s(s + a)(s + b); this is unstable with large K but can become stable if K is reduced, the point at which the plot crosses the axis being –K/(a + b) and so stability is when –K/(a + b) > –1*

Example

Plot the Nyquist diagram for a system with the open-loop transfer function $K/(s + 1)(s + 2)(s + 3)$ and consider the value of K needed for stability.

The open-loop frequency response is:

$$\frac{K}{(j\omega + 1)(j\omega + 2)(j\omega + 3)}$$

The magnitude and phase are:

$$\text{magnitude} = \frac{K}{\sqrt{(\omega^2 + 1)(\omega^2 + 4)(\omega^2 + 9)}}$$

$$\text{phase} = \tan^{-1}\left(\frac{\omega}{1}\right) + \tan^{-1}\left(\frac{\omega}{2}\right) + \tan^{-1}\left(\frac{\omega}{3}\right)$$

When $\omega = 0$ then the magnitude is $K/6$ and the phase is 0°. When $\omega = \infty$ then the magnitude is 0 and the phase is 270°. We can use these, and other points to plot the polar graph.

Alternatively we can consider the frequency response in terms of real and imaginary parts. We can write the open-loop frequency function as:

$$\frac{6K(1-\omega^2)}{(\omega^2+1)(\omega^2+4)(\omega^2+9)} + j\frac{\omega K(\omega^2-11)}{(\omega^2+1)(\omega^2+4)(\omega^2+9)}$$

When $\omega = 0$ then the imaginary part is zero and the real part is $K/6$. When $\omega = \infty$ then the imaginary part is zero and the real part is 0. The imaginary part will be zero when $\omega = \sqrt{11}$. This is a real part, and hence magnitude, of $-K/60$ and is the point at which the plot crosses the real axis. Thus for a stable system we must have $-K/60$ less than -1, i.e. K must be less than 60. Figure 6.11 shows the complete Nyquist plot (not to scale).

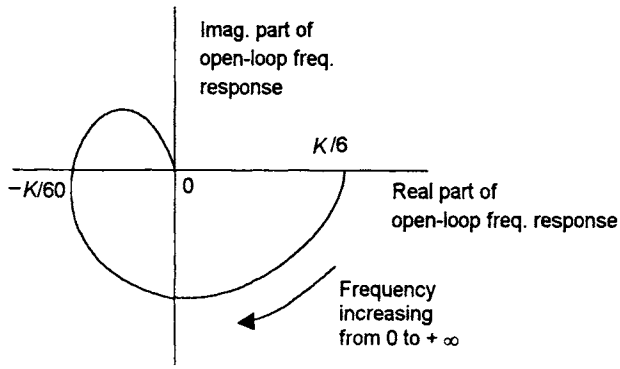

Figure 6.11 *Example*

6.4 Relative stability

The use of gain margin and phase margin was introduced in Section 5.6.1 to discuss the relative stability of a system in the frequency domain when described by a Bode plot. With Nyquist plots:

1 *Gain margin*

The phase crossover frequency is the frequency at which the phase angle first reaches $-180°$ and thus is the point where the Nyquist plot crosses the real axis (Figure 6.12). On a Nyquist plot the $(-1, j0)$ point is the point separating stability from instability. The gain margin is the amount by which the actual gain must be multiplied before the onset of instability. Thus if the plot cuts the negative real axis at $-x$ (Figure 6.12), it has to be multiplied by $1/x$ to give the value -1 and so the gain margin, which is expressed in dB, is $20 \lg(1/x)$.

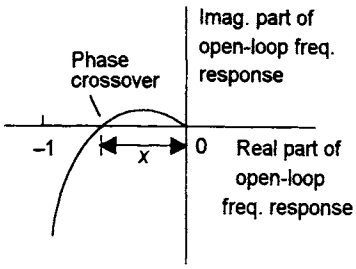

Figure 6.12 *Phase crossover and gain margin*

When the open-loop plot goes through the $(-1, j0)$ point the gain margin is 0 dB, the system being on the margin of instability. When the open-loop plot goes to the left of $(-1, j0)$ point the gain margin is negative in dB, the system being unstable. When the open-loop plot goes to the right of $(-1, j0)$ point the gain margin is positive in dB,

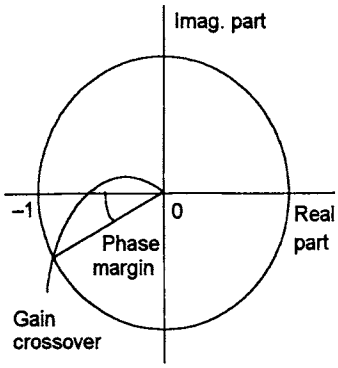

Figure 6.13 *Phase margin: the angle through which the gain crossover line must be rotated to reach the real axis and pass through the (–1, j0) point*

the system being stable. When the open-loop plot does not intersect the negative real axis the gain margin is infinite in dB.

2 *Phase margin*

The phase margin is defined as the angle in degrees by which the phase angle is smaller than –180° at the gain crossover, the gain crossover being the frequency at which the open-loop gain first reaches 1. Thus, with a Nyquist plot, if we draw a circle of radius 1 centred on the origin, then the point at which it intersects the Nyquist line gives the gain crossover. The phase margin is the angle through which this gain crossover line must be rotated about the origin to reach the real axis and pass through the (–1, j0) point (Figure 6.13).

Example

Determine the gain margin and the phase margin for a system with the open-loop transfer function $K/(s + 1)(s + 2)(s + 3)$ with $K = 20$. This system was discussed earlier in this chapter (see Figure 6.11 for the Nyquist plot).

The open-loop frequency response is:

$$\frac{K}{(j\omega + 1)(j\omega + 2)(j\omega + 3)}$$

and this can be rearranged to give:

$$\frac{6K(1 - \omega^2)}{(\omega^2 + 1)(\omega^2 + 4)(\omega^2 + 9)} + j\frac{\omega K(\omega^2 - 11)}{(\omega^2 + 1)(\omega^2 + 4)(\omega^2 + 9)}$$

The imaginary part will be zero when $\omega = \sqrt{11}$ and thus the real part is $-K/60$ and is the point at which the plot crosses the real axis. Hence, if we have $K = 20$ then the plot intersects the negative real axis at $-20/60 = -1/3$. The gain can thus be increased by a factor of 3 in order to reach the –1 point. The gain margin is thus 20 lg 3 = 9.5 dB.

The magnitude is:

$$\text{magnitude} = \frac{K}{\sqrt{(\omega^2 + 1)(\omega^2 + 4)(\omega^2 + 9)}}$$

Thus, for $K = 20$, the magnitude is 1 when $\omega = 1.84$ rad/s. The phase is given by:

$$\text{phase} = \tan^{-1}\left(\frac{\omega}{1}\right) + \tan^{-1}\left(\frac{\omega}{2}\right) + \tan^{-1}\left(\frac{\omega}{3}\right)$$

and so, at this frequency, the phase is –135.5°. Thus the phase margin is 44.5°.

Problems

1 Sketch the Nyquist diagram for a system having an open-loop transfer function of $1/[s(s+1)]$.

2 With a Nyquist diagram for the open-loop frequency response for a system, what is the condition for the system to be stable?

3 Determine the gain margin and the phase margin for a system which gave the following open-loop frequency response:

Freq. rad/s	1.4	2.0	2.6	3.2	3.8
Magnitude	1.6	1.0	0.6	0.4	0.2
Phase deg.	−150	−160	−170	−180	−190

4 Determine the gain margin and the phase margin for a system which gave the following open-loop frequency response:

Freq. rad/s	4	5	6	8	10
Gain	3.2	2.3	1.7	1.0	0.6
Phase in deg.	−140	−150	−157	−170	−180

5 Determine the gain margin and the phase margin for a system having an open-loop transfer function of:

$$\frac{K}{s(s+1)(s+2)}$$

when $K = 4$.

6 Determine the gain margin and the phase margin for a system having an open-loop transfer function of :

$$\frac{1}{s(0.2s+1)(0.05s+1)}$$